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YouTube Comment Sentiment Analysis

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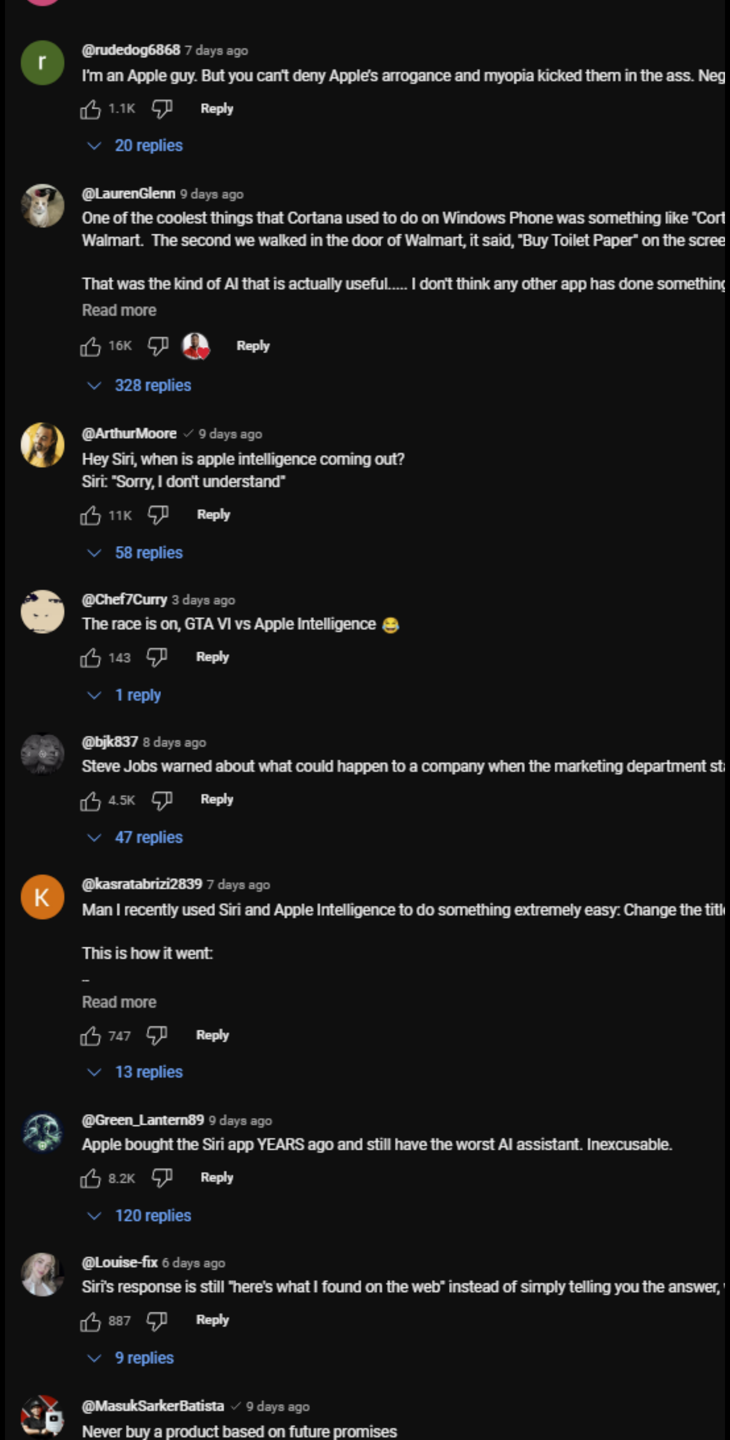


Introduction

- Text classification to classify YouTube comments as positive, negative, or neutral

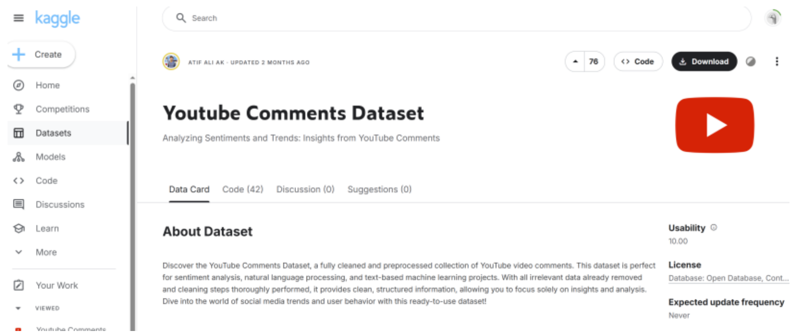
- Benefits user engagement and detecting harmful content

- Challenges include an imbalanced dataset, emojis, slang, special characters

- Provide content creators with insights into viewer reactions

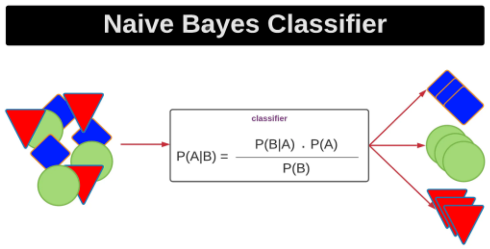
Dataset

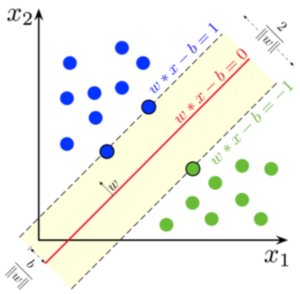
- Contained ~18,000 unique comments
- Was imbalanced with the following distribution:
  - Positive – 62%
  - Negative – 13%
  - Neutral – 25%
- To combat this, we used SMOTE for a sampling technique

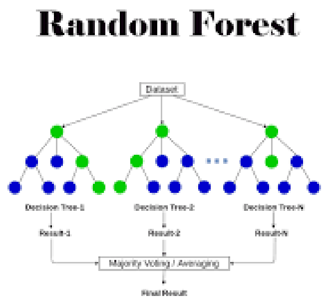


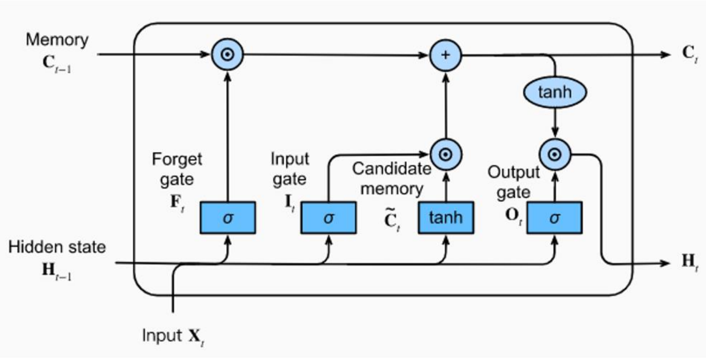
Literature Review

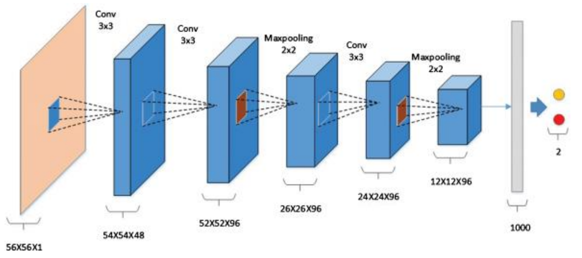
- Earlier studies explored use of:
  - Naive Bayes
  - Support Vector Machines
  - Random Forest
- More recent studies have used:
  - LSTMs
  - CNNs
  - Transformers

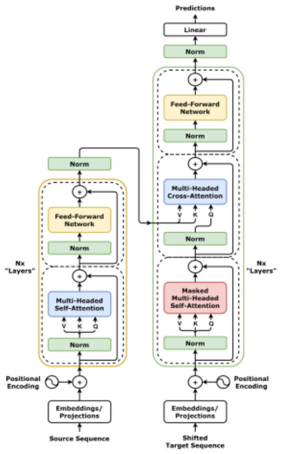












Methodology

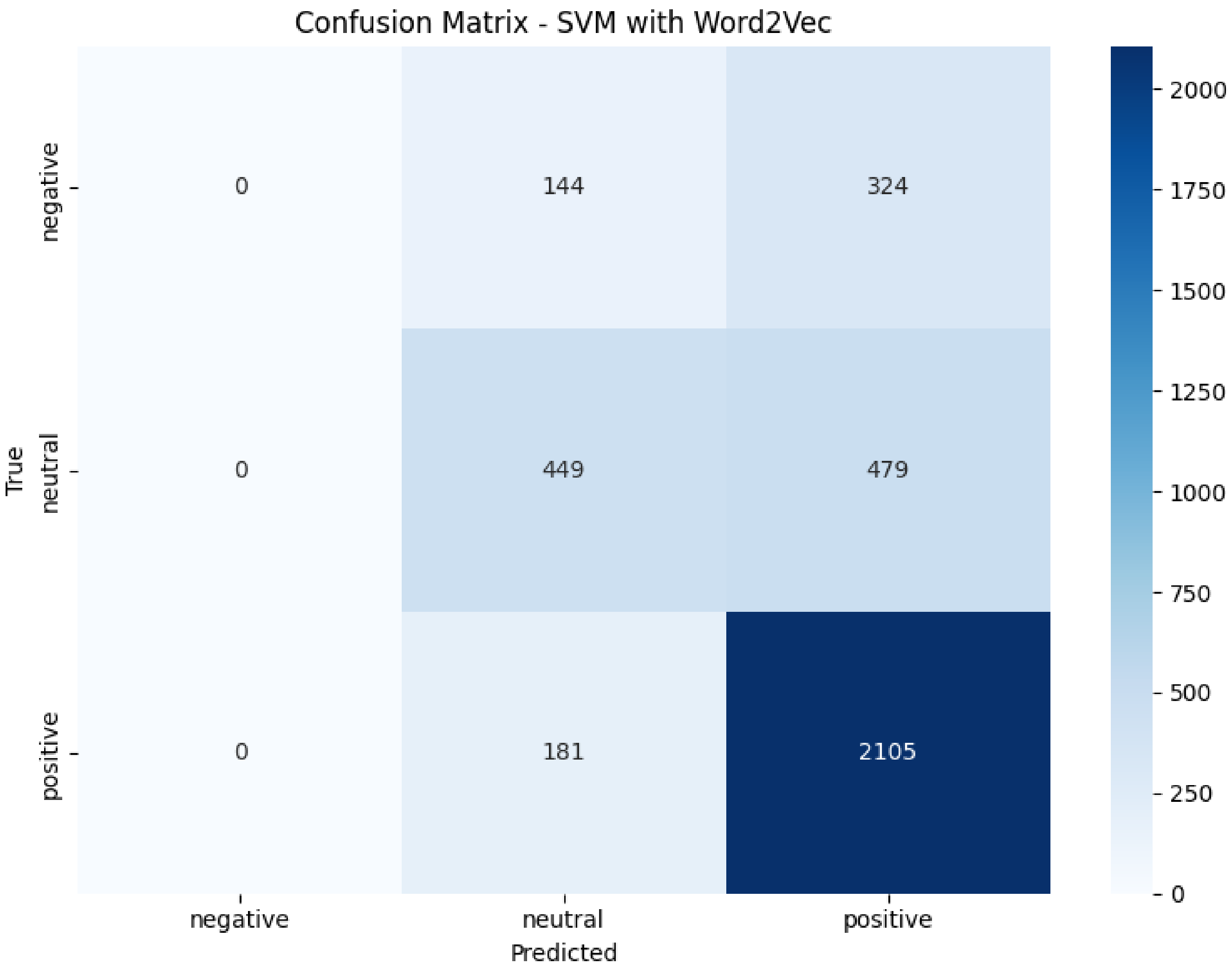
- Planned to use traditional machine learning methods
  – NB, SVM, and RF
- Also include advanced methods
  – SVM with Word2Vec
  – SVM with Co-occurrence SVD
- Performance evaluators
  – Accuracy, precision, recall, f1 score

Experiments

- Performance metric table

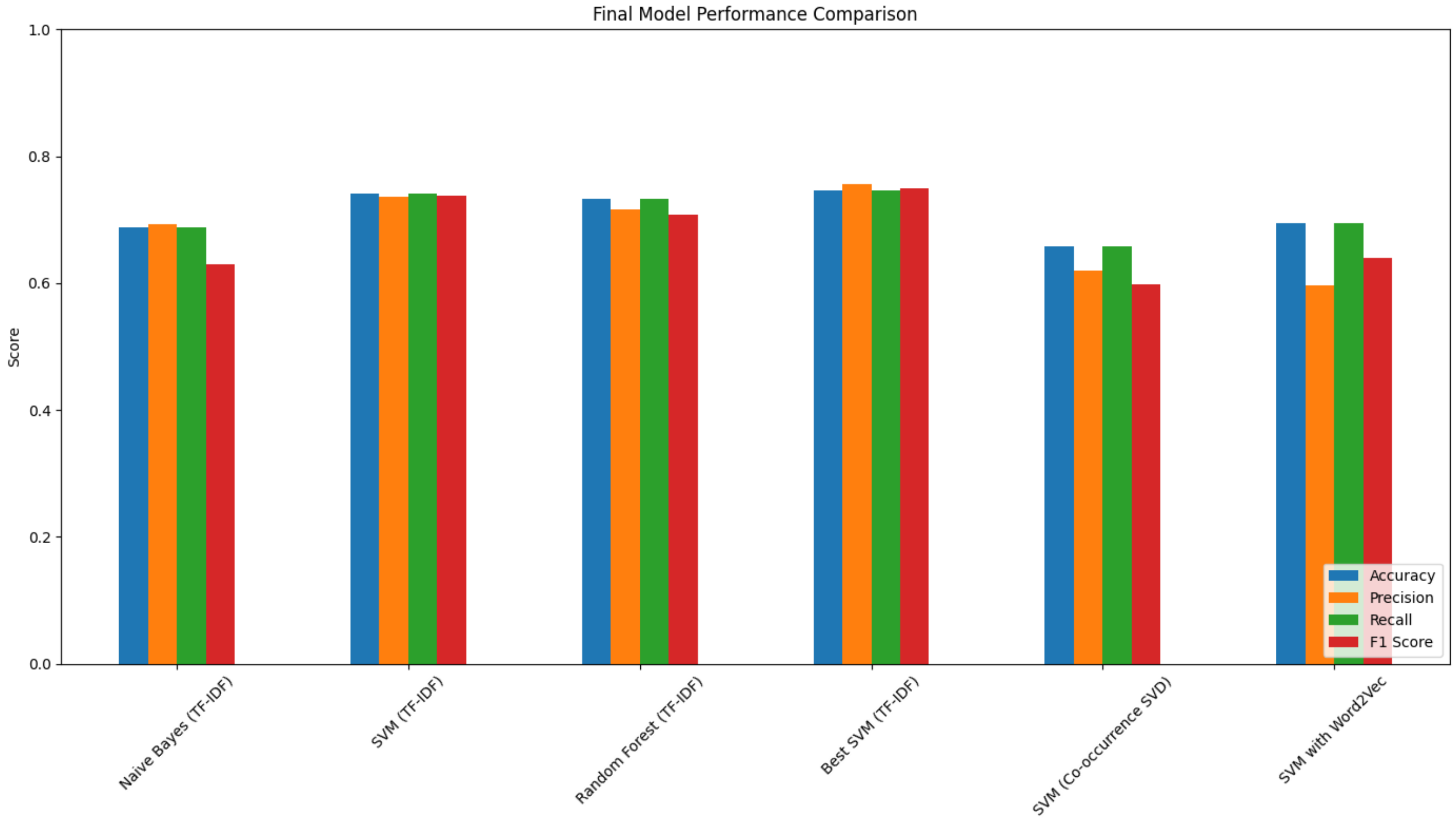
Model	Accuracy	Precision	Recall	F1 Score
Naive Bayes (TF-IDF)	0.6874	0.6927	0.6874	0.6303
SVM (TF-IDF)	0.7412	0.7359	0.7412	0.7379
SVM (Word2Vec)	0.6942	0.5963	0.6942	0.6376
SVM (Co-occurrence SVD)	0.6583	0.6204	0.6583	0.5975
Best SVM (TF-IDF)	**0.7452**	**0.7567**	**0.7452**	**0.7501**
RF (IF-IDF)	0.7333	0.7166	0.7333	0.7070

Experiments



Experiments

- Performance metric graph



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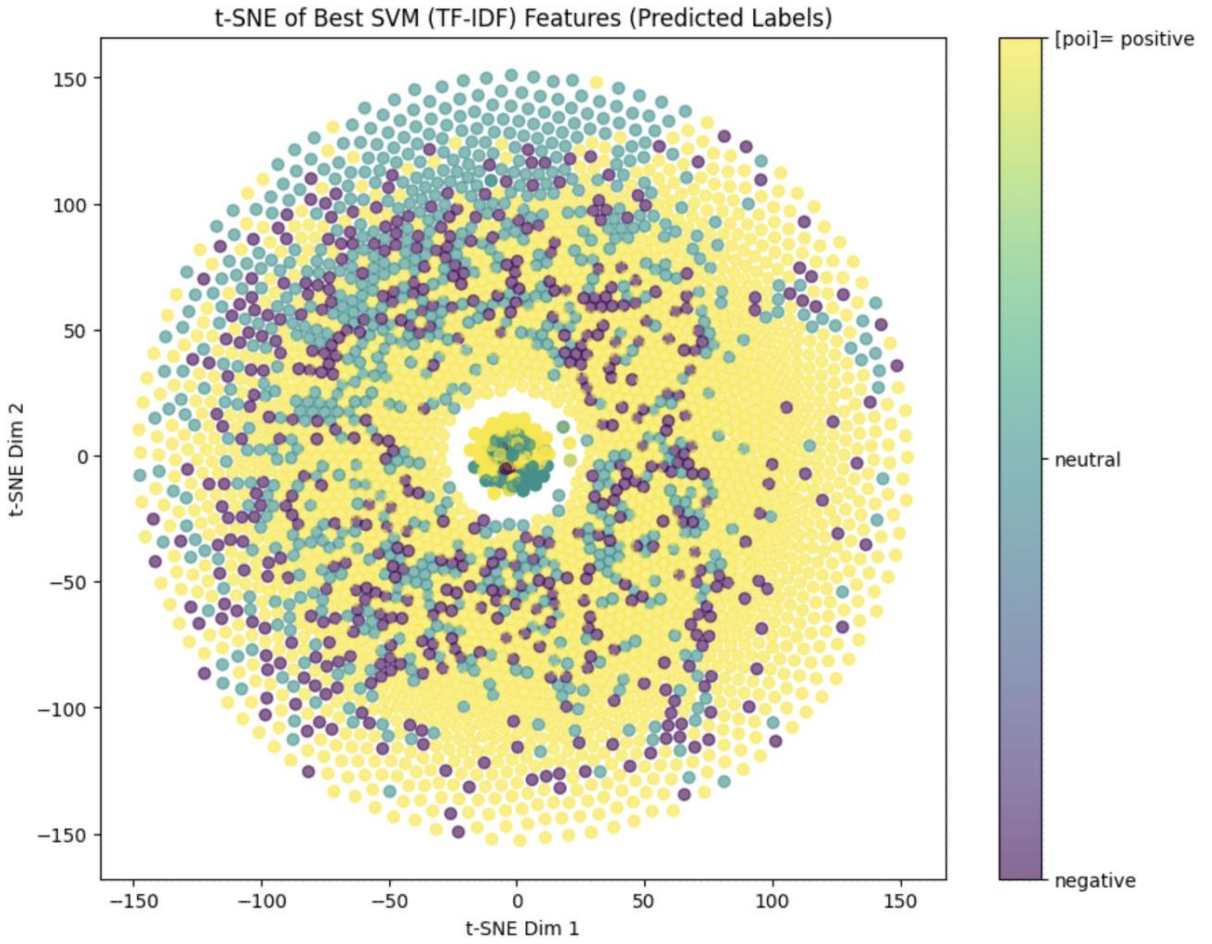
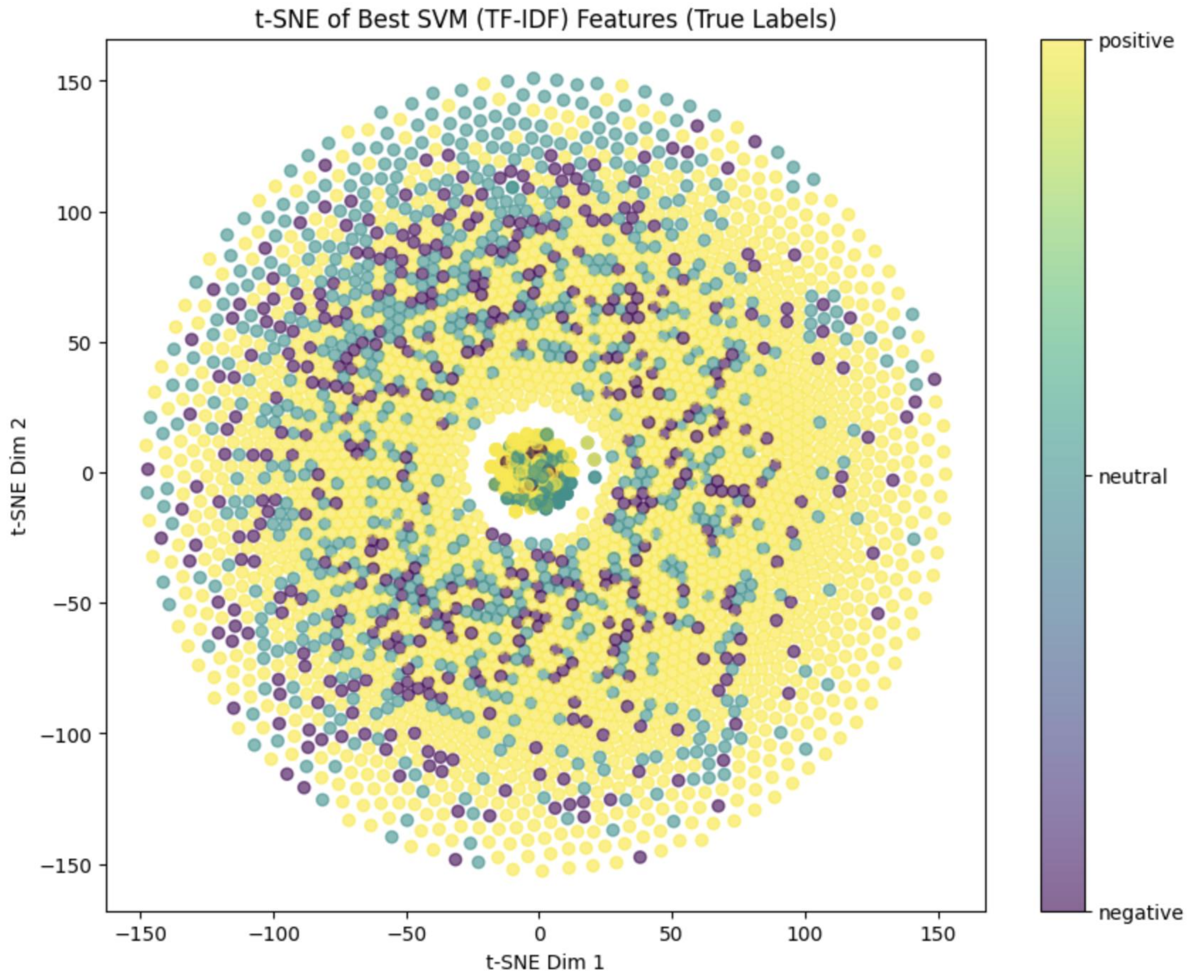
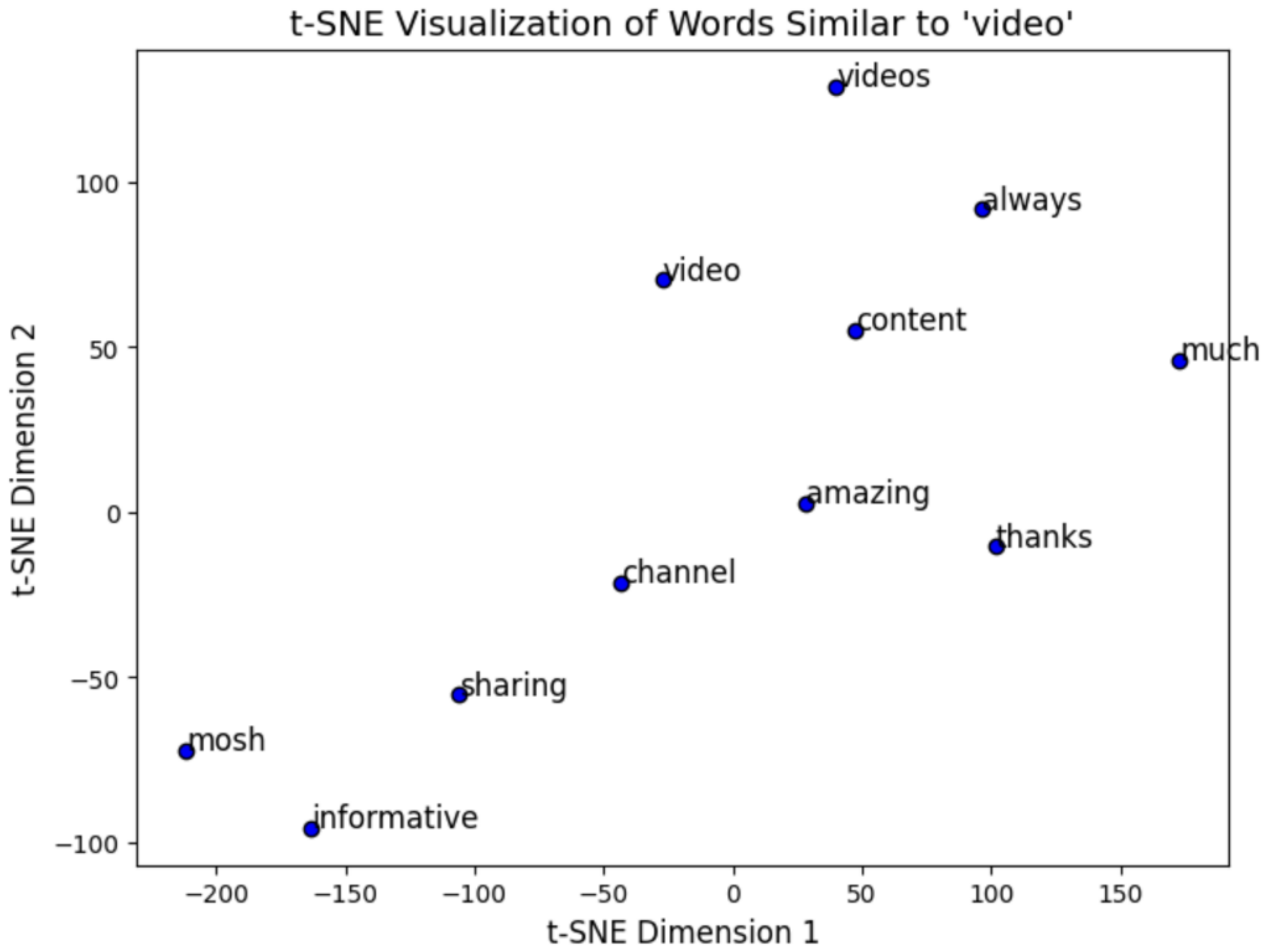
Experiments





Experiments



Experiments

- Misclassifications
  - We observed that foreign languages are often counted positive
  - Even humans manually annotating these would have a hard time

Comment	True Sentiment	Predicted Sentiment
Imagine scammer get hurt someone points win	negative	positive
Bihar ba	neutral	positive
Watching asmr ramadan	neutral	positive
Didn’t believe anything wrong refusing turn doctor	negative	neutral
bài của buffalove thực sự quá hay đi ạ em	neutral	positive
Since little knew I'd rather live street become	negative	neutral
You're friends zhc interesting	neutral	positive

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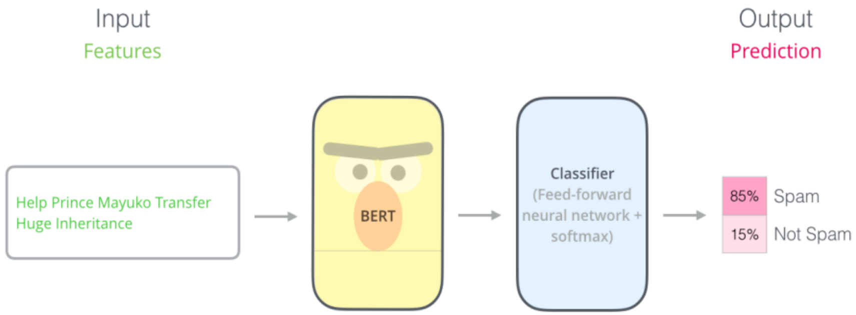
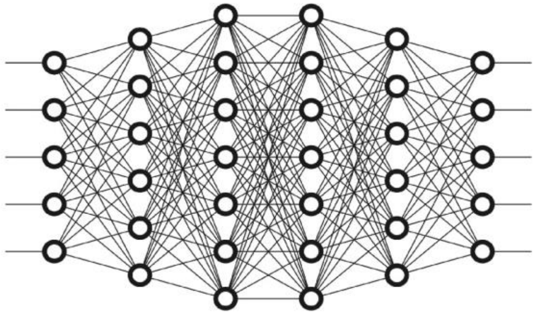
Conclusion

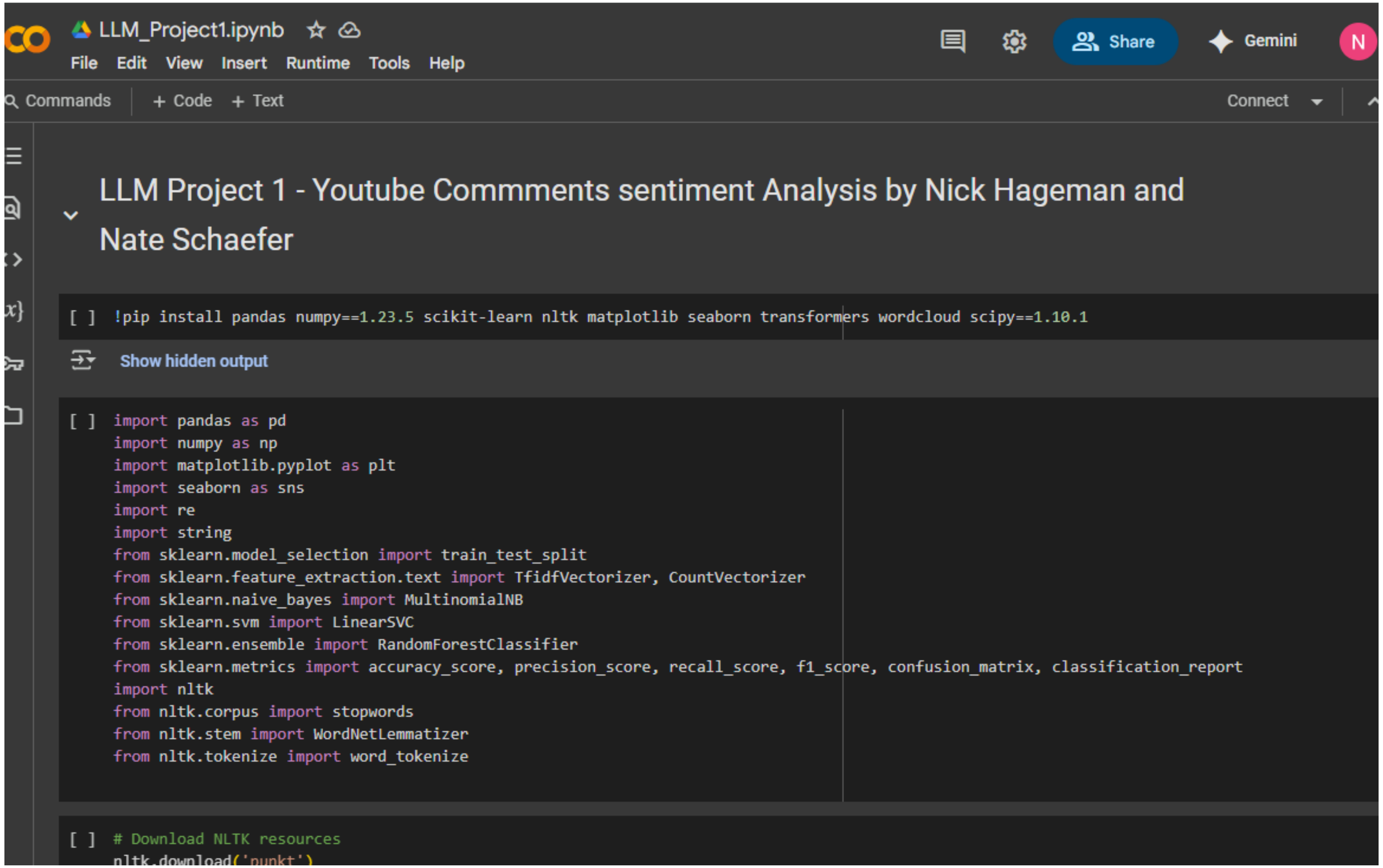
- SVM model with hyperparameter tuning performed the best in all performance metrics
- Future work could include incorporating more pre-trained models such as BERT



- Also could experiment with other deep learning models too



Code Overview



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